

# NyayaSathi

India's AI Legal Helpline

Prepared for Saurabh Karn, Founding Member — Sarvam AI

<5s

End-to-End

11

Languages

50+

Legal Topics

26

Acts Validated

24/7

Free Service

## What is NyayaSathi?

A **voice-first AI legal assistant** built for rural India. Users speak their legal problem in any of 11 Indian languages — the system listens, understands, retrieves relevant law, generates a verified legal response, and speaks it back. Works on **web browsers** (any phone with Chrome) and **IVR phone calls** (Exotel toll-free) for feature phone users with no internet. Completely free, no login, no app download.

## Technical Architecture

- **LLM Race Engine:** Gemini Flash + Sarvam-105B run concurrently via Promise.race. Sarvam starts 400ms delayed (Gemini head start). First valid response wins. Fallback timeout at 6s. Gives speed of Gemini with Indic depth of Sarvam.
- **Voice Pipeline:** Sarvam Bulbul v3 TTS (11 languages, male/female voices) + Sarvam STT. Audio chunked at 150 chars for streaming — first audio plays in ~2s. Frontend streams NDJSON chunks.
- **3-Tier BM25 RAG:** Supreme Court corpus (landmark judgments, fundamental rights), High Court corpus (RERA, bail, writs), District Court corpus (FIR, rent, fraud, rural issues). Cross-lingual retrieval via 65+ Hindi colloquial-to-legal term mappings (e.g., dahej→dowry, peshi→court hearing).
- **10-Rule Post-LLM Pipeline:** stripMarkdown → enforceLanguage → verifyCitations → detectEmergency → ensureNALSA → enforceWordLimit → filterNonLegal → convertNumbers → filterProfanity → guardHallucination. Every response passes through all 10 rules before reaching the user.
- **Anti-Hallucination:** Citation grounding score (0.7 threshold) verifies section numbers against RAG chunks. ACT\_SECTION\_RANGES validates 26 Indian Acts — hallucinated sections (e.g., 'IPC Section 900') are stripped from the response.

## Built for Rural India

- **Rural Corpus (15 chunks):** Land boundary disputes, Khatauni/Khasra corrections, Panchayat corruption, PMFBY crop insurance, PM-KISAN/NREGA/Ayushman eligibility, Aadhaar corrections, ration card issues, stalking (BNS 78), digital arrest scams, WhatsApp/UPI fraud, SIM swap attacks, caste discrimination (SC/ST Act), reservation denial, widow/old-age pension, child/forced marriage.
- **Emergency Detection:** Detects domestic violence, suicide, child abuse keywords → auto-appends helpline numbers: Women 181, Cyber 1930, NALSA 15100, Police 112.
- **Cross-Lingual Retrieval:** 65+ Hindi colloquial mappings — 'dahej' retrieves dowry law, 'tareekh' retrieves court date info, 'batwara' retrieves property partition law. Rural users don't need to know legal terminology.
- **IVR Phone Support:** Exotel toll-free number for feature phone users. Full voice interaction without internet. Same AI pipeline as web.

## Stack

**LLM:** Google Gemini Flash + Sarvam-105B | **Voice:** Sarvam Bulbul v3 TTS/STT | **Backend:** Node.js + Express | **Phone:** Exotel IVR | **RAG:** BM25 three-tier | **Languages:** Hindi, English, Bengali, Telugu, Tamil, Marathi, Gujarati, Kannada, Malayalam, Punjabi, Odia

## Discussion Points for Saurabh

1. How did OpenNyAI/Jugalbandi handle the gap between formal statutory language and colloquial Hindi queries?
2. Recommended Sarvam TTS chunk size for streaming? We optimized 300→150 chars — is there a streaming endpoint planned?
3. BM25 vs dense embeddings for Indian legal retrieval — did you hit ceiling issues at Samta Law?
4. Sarvam STT accuracy for rural dialects (Bhojpuri-Hindi, tribal) — any accent adaptation roadmap?
5. Where does legal AI fit in Sarvam's product roadmap — vertical or horizontal infrastructure play?